

Department of Automobile Engineering

Basic Automobile Engineering

Course Code: 2051

Semester: 2

Revision: 2021

Credits: 3

A comprehensive guide to the fundamentals of vehicle classification, engine working, and constructional details.

Course Overview

Category	Engineering Science	Periods/Week	3 (L:3, T:0, P:0)
Total Periods	45 per semester	Assessment	CIE (50) & ESE (50)

Course Objectives

- To know the history and classification of automobiles.
- To understand various Vehicle Chassis Terminology.
- To appreciate the working and construction of different engines/components.
- To classify different classes of vehicles based on usage, axle weight, and power plant type.

Course Outcomes (COs)

CO No.	Description	Hours	Cognitive Level
CO1	Summarize different classification of vehicles and terminology of Chassis specification.	10	Understanding
CO2	Make use of Petrol and Diesel Engines.	11	Applying
CO3	Illustrate the working and constructional details of various engine components.	11	Understanding

CO No.	Description	Hours	Cognitive Level
CO4	Outline the constructional details of camshaft, valves and valve operating mechanism.	11	Understanding

Module 1: History and Classification

Duration: 10 Hours | CO1

1.1 History and Development

The automobile has evolved from steam-powered "road locomotives" to modern electric and autonomous vehicles. Key milestones include:

- Abroad:** Nicholas Cugnot (1769 steam carriage), Karl Benz (1886 - first practical petrol car), Henry Ford (Mass production via Model T).
- India:** Pre-independence saw imports. Post-independence (1947) led to the establishment of Hindustan Motors (Ambassador) and Premier Automobiles (Padmini). The 1980s revolution started with Maruti Suzuki.

Malayalam support:

ഓട്ടോമൊബൈൽ വാഹനം 1769-ൽ നിക്കോളാസ് കുഗ്നോട്ട് ഓട്ടോമൊബൈൽ വാഹനം വികസിപ്പിച്ചു. കാൾ ബെൻസ് (1886) ആദ്യത്തെ പ്രായോഗിക പെട്രോൾ കാർ വികസിപ്പിച്ചു. ഹെൻറി ഫോർഡ് (മോഡൽ ടി) വാഹനം массовое производство. സ്വാതന്ത്ര്യത്തിന് ശേഷം (1947) ഇന്ത്യയിൽ ഹിന്ദുസ്ഥാൻ മോട്ടോർസ് (അംബാസാദർ) പ്രീമിയർ ഓട്ടോമൊബൈൽസ് (പാമിനി) സ്ഥാപിതമായി. 1980-കളുടെ വിപ്ലവം മാരുതി സുസുക്കിയിൽ നിന്ന് ആരംഭിച്ചു.

1.2 Classification of Vehicles

Automobiles are classified based on several criteria:

- Purpose:** Passenger (Car, Bus), Goods (Truck, Lorry).
- Capacity:** LMV (Light Motor Vehicle), HTV (Heavy Transport Vehicle).
- Fuel:** Petrol, Diesel, CNG, Electric, Hybrid.
- Axes:** Two-wheeler, Three-wheeler, Four-wheeler, Multi-axle.
- Drive:** Left-hand drive, Right-hand drive.

1.3 Chassis Terminology & Layouts

The **Chassis** is the skeleton of the vehicle, carrying all major systems like the engine, transmission, and suspension.

Key Specifications

Term	Definition
Wheel Base	Distance between the centers of front and rear wheels.

Wheel Track	Distance between the centers of the left and right wheels on the same axle.
Ground Clearance	Distance between the lowest point of the vehicle and the ground.
Kerb Weight	Weight of the vehicle with full fuel and equipment, but without passengers/load.
Gross Weight	Maximum allowable weight (Kerb weight + Payload).

Module 2: Engine Terminology & Working

Duration: 11 Hours | CO2

2.1 Heat Engines: IC vs EC

A Heat Engine converts thermal energy into mechanical work.

- External Combustion (EC):** Fuel burns outside the cylinder (e.g., Steam Engine).
- Internal Combustion (IC):** Fuel burns inside the cylinder (e.g., Petrol/Diesel Engine).

2.2 Engine Terminology

- Bore:** Inner diameter of the cylinder.
- Stroke:** Distance traveled by the piston from TDC to BDC.
- TDC (Top Dead Center):** Highest point the piston reaches.
- BDC (Bottom Dead Center):** Lowest point the piston reaches.
- Swept Volume (V_s):** Volume displaced by the piston in one stroke.
- Clearance Volume (V_c):** Volume above the piston at TDC.

2.3 4-Stroke vs 2-Stroke Engines

In a **4-stroke engine**, one power stroke occurs every two revolutions of the crankshaft (Suction, Compression, Power, Exhaust). In a **2-stroke engine**, one power stroke occurs every single revolution.

Malayalam support:

4-സ്റ്റോക്ക് എഞ്ചിൻ (Suction, Compression, Power, Exhaust) രണ്ട് റിവോളൂഷനുകൾക്ക് ഒരു പവർ സ്റ്റോക്ക് ഉണ്ടാകുന്നു. 2-സ്റ്റോക്ക് എഞ്ചിൻ ഒരു റിവോളൂഷനിലെ ഒരു പവർ സ്റ്റോക്ക് ഉണ്ടാകുന്നു.

Formula Bank & Calculations

$$\text{Compression Ratio (CR)} = (V_s + V_c) / V_c$$

Example Calculation:

Given: Swept Volume (V_s) = 450cc, Clearance Volume (V_c) = 50cc.

Calculation: $CR = (450 + 50) / 50 = 500 / 50 = 10$.

Answer: The Compression Ratio is 10:1.

Compression Ratio Calculator

Swept Volume (cc):

e.g. 150

Clearance Volume (cc):

e.g. 15

Calculate CR

$$\text{Mechanical Efficiency } (\eta_m) = (\text{Brake Power} / \text{Indicated Power}) \times 100$$

Visual Library

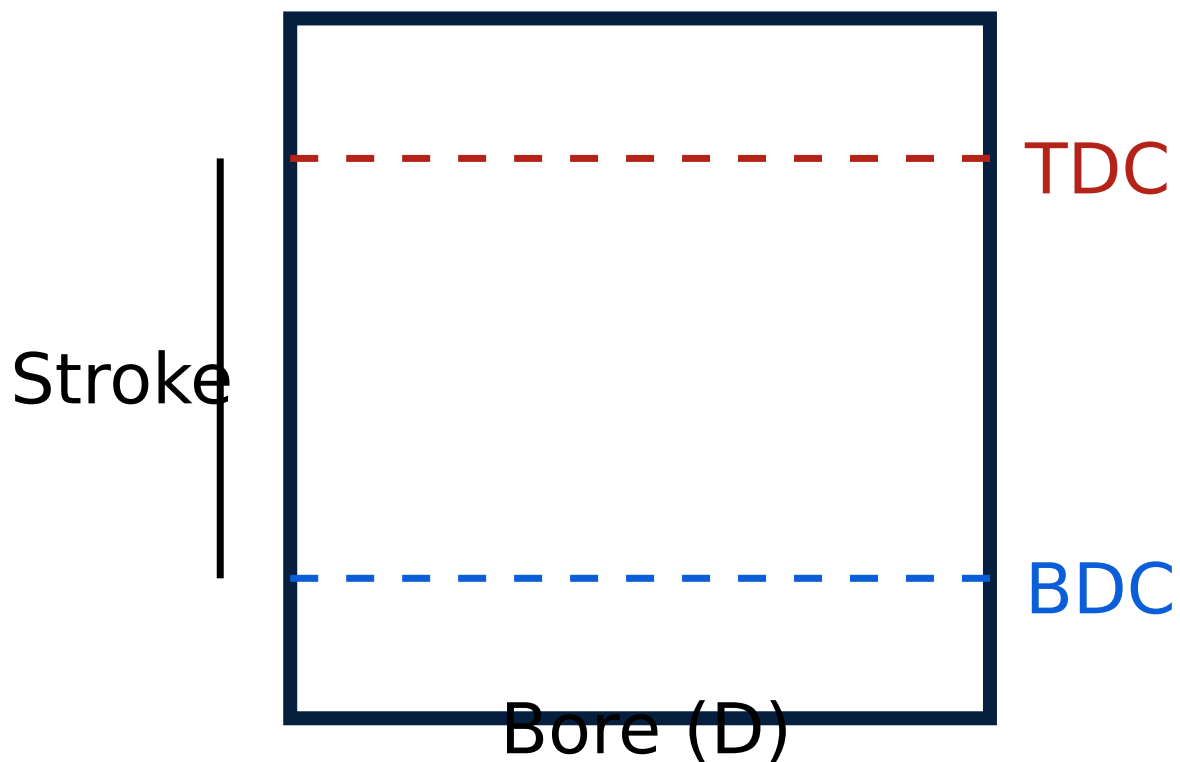


Figure 1: Basic Engine Cylinder Terminology (Bore, Stroke, TDC, BDC)

Module 3: Engine Construction Details - I

Duration: 11 Hours | CO3

3.1 Cylinder Block and Head

Cylinder Block: The main body of the engine. Usually made of Cast Iron or Aluminum Alloy. It contains the cylinders and water jackets for cooling.

Cylinder Head: Closes the top of the cylinder. Houses valves, spark plugs (SI), or injectors (CI).

3.2 Cylinder Liners

Liners are sleeves inserted into the block to provide a wear-resistant surface.

- **Dry Liner:** Not in direct contact with coolant. Easy to replace but poor heat transfer.
- **Wet Liner:** Direct contact with coolant. Better cooling but requires sealing to prevent leaks.

3.3 Piston and Rings

The piston transmits the force of combustion to the connecting rod. Made of Aluminum alloy for lightness.

Piston Rings: 1. Compression Rings (Seal the combustion chamber). 2. Oil Control Rings (Scrape excess oil from cylinder walls).

Module 4: Engine Construction Details - II

Duration: 11 Hours | CO4

4.1 Firing Order

The sequence in which power strokes occur in different cylinders. It ensures engine balance and smooth torque.

- 3-Cylinder: 1-2-3 or 1-3-2
- 4-Cylinder: 1-3-4-2 or 1-2-4-3
- 6-Cylinder: 1-5-3-6-2-4

4.2 Valve Operating Mechanism

Valves control the flow of air/fuel and exhaust. Most modern engines use **Poppet Valves**.

- **SOHC (Single Overhead Camshaft):** One cam operates both intake and exhaust valves.
- **DOHC (Double Overhead Camshaft):** Separate cams for intake and exhaust, allowing better performance at high RPM.

Malayalam support:

വലുതായ എഞ്ചിനുകളിൽ ഉപയോഗിക്കുന്ന വാൽവ് ഓപ്പറേറ്റിംഗ് മെക്കാനിസം (Camshaft) രണ്ടുതരത്തിലുണ്ട്. SOHC-ൽ ഒരു വാൽവ് ഓപ്പറേറ്റിംഗ് മെക്കാനിസം ഉപയോഗിക്കുന്നു. DOHC-ൽ രണ്ടു വാൽവ് ഓപ്പറേറ്റിംഗ് മെക്കാനിസം ഉപയോഗിക്കുന്നു.

Module-wise Practice Questions

Q1: Define Wheelbase. (1 Mark)

Wheelbase is the distance between the centers of the front and rear wheels of a vehicle.

Q2: Differentiate between Petrol and Diesel engines. (5 Marks)

Feature	Petrol Engine (SI)	Diesel Engine (CI)
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Fuel	Petrol	Diesel
Ignition	Spark Plug	Compression Heat
Compression Ratio	Lower (6-10)	Higher (14-22)
Efficiency	Lower	Higher

Q3: Explain the function of Piston Rings. (3 Marks)

Piston rings serve three main purposes: sealing the combustion chamber to prevent gas leakage, transferring heat from the piston to the cylinder wall, and controlling the oil film on the cylinder wall.

Practice Model Paper

(Based on REV2021 Pattern - Not an official paper)

Part A (Short Answer - 2 Marks each)

1. Define Kerb Weight.
2. What is TDC and BDC?
3. List two materials used for cylinder blocks.

Part B (Medium Answer - 5 Marks each)

1. Explain the classification of vehicles based on drive.
2. Compare 2-stroke and 4-stroke engines.
3. Explain the need for firing order in multi-cylinder engines.

Part C (Long Answer - 10 Marks each)

1. With a neat sketch, explain the working of a 4-stroke petrol engine.
2. Describe the constructional details of a piston and explain expansion control methods.

Quick Revision

- **Chassis:** The backbone of the vehicle.
- **IC Engine:** Combustion happens inside (Petrol/Diesel).
- **CR:** Ratio of total volume to clearance volume.
- **Liners:** Dry (no water contact) vs Wet (water contact).
- **DOHC:** Better for high-speed performance.

Glossary

Bore

Diameter of the engine cylinder.

Stroke

Distance moved by piston between TDC and BDC.

Payload

The weight of passengers and cargo a vehicle can carry.

Gasket

A seal used between two surfaces (like block and head) to prevent leaks.

Official Source Declaration

This content is based on the **SITTTR Kerala REV2021 Syllabus** for Course Code **2051 (Basic Automobile Engineering)**. The primary source is the official curriculum document provided by the State Institute of Technical Teachers' Training and Research, Kerala.

Reference Books listed in syllabus: Kirpal Singh (Vol II), R.B. Gupta, William H. Crouse.

മലയാളം അഭ്യർത്ഥന / Malayalam learning support

ഈ ഉള്ളടക്കം സി.ടി.ടി.ടി.ആർ. കേരളം റിവ്യൂ 2021 സിലബസ് പ്രകാരം തയ്യാറാക്കിയതാണ്. പ്രാഥമിക സ്രോതസ്സ് കേരള സംസ്ഥാന ടെക്നിക്കൽ ടീച്ചേഴ്സ് ട്രെയിനിംഗ് റിസർച്ച് ഇൻസ്റ്റിറ്റ്യൂട്ടിന്റെ ഔദ്യോഗിക കർമ്മസൂചിയാണ്. സിലബസിൽ പരാമർശിച്ചിരിക്കുന്ന പുസ്തകങ്ങൾ: കിര്പാൽ സിംഗ് (വോൾ II), എസ്.ആർ. ഗുപ്ത, വില്യം എച്ച്. ക്രൗസ്.